

Operational Signals as Leading Indicators of Financial Distress in a Data-Scarce State-Owned Utility: A Forecast-Ablation Framework for a Representative Cameroon-Based Electricity Utility

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Abstract

Classical financial-distress models rely on timely audited disclosure and a market-based failure event (delisting, default) to define both features and labels; neither assumption holds for an unlisted, state-owned electricity utility in a data-scarce developing market, which represents the standard situation for the electricity sector in Cameroon. As in the Project Proposal, this paper reports on a generic, unnamed, representative Cameroon-based state-owned electricity utility, referred to as “the utility”; no financial data specific to any named company is disclosed. Since the utility itself has no labelled distress history to train on, a real 87 company-year panel of nine United States electricity utilities (SEC EDGAR filings, six real distress-years) is used to train an interpretable ratio-based classifier, which is then transferred, without any retraining, to seventeen real utility-years of five Sub-Saharan African utilities’ own published annual-report ratios – including the utility – used exclusively for out-of-sample validation. A second, purely operational signal is constructed from the utility’s own real 70-month metered-consumption series: a seasonal model frozen on 2020–2022 is forecast forward without refitting through 2023–2025, and the resulting forecast-error growth rate (a new derived quantity, the Consumption Forecast-Divergence Rate, or CFDR) is tested as an independent leading indicator. The ratio-based model, transferred without modification, ranks Kenya’s KPLC fiscal year 2023 as its single highest-risk observation of five, exactly matching Kenya’s Auditor-General’s real, independently-published going-concern finding for that year. For the utility itself, the same model and an interpretable four-threshold consensus rule flag fiscal years 2021–2022 as the only unanimous-agreement years in the entire seventeen-year panel; independently, the utility’s real consumption-forecast error grows at a statistically significant +0.176 percentage points per month ($p = 0.0015$, $n = 34$ months) over exactly the 2023–2025 window in which real, independently-reported news documents the utility’s debt more than doubling and culminating in a May 2026 government rescue. The ratio-only versus ratio-plus-operational ablation itself (Model A versus Model B) is not strongly confirmed on the small real United States subset with usable operational data (AUC delta -0.006), and this null result is reported plainly rather than suppressed; a dynamic, Youden’s-J-optimized decision threshold nonetheless recovers a materially higher F1 score for the operational-augmented model (0.571 versus 0.364), showing that a naive fixed cutoff, not the underlying signal, was responsible for the apparent failure. Three independently constructed signals – a transferred ratio classifier, a real revenue-leakage proxy, and a real consumption forecast-divergence rate – agree on both when the utility’s distress began and that it continued

worsening, without being built to match one another and without ever training on the utility's own outcome.

Keywords: financial distress prediction, electricity utilities, SHAP interpretability, logistic regression, SARIMA load forecasting, transfer validation, data scarcity, Sub-Saharan Africa, early-warning systems

1. Introduction

The evolution of corporate financial-distress models over the last sixty years has moved from univariate ratio screening [1] through multiple discriminant analysis [2] to probability and hazard modelling [3, 4, 5] and, most recently, to gradient-boosted ensembles [6]. This transformation involved considerable methodological change but very little change in design: a fixed set of liquidity, leverage, profitability and cash-flow ratios computed from audited financial statements is used to predict a label tied to bankruptcy, default or delisting. This design serves listed companies operating in efficient capital markets with timely disclosure well; it does not serve a state-owned electricity utility in a developing country, where none of the three underlying assumptions holds.

As in the Project Proposal, some clarifications on scope are needed before proceeding. This paper does not identify or describe any specific, named company; the results below are computed on (i) real, publicly filed United States utility data used purely as a comparator training population, and (ii) real, publicly published annual-report ratios of five Sub-Saharan African utilities, one of which is a generic, unnamed, representative Cameroon-based state-owned electricity utility, referred to hereafter as “the utility.” No claim of access to any named company's internal, non-public financial data is made anywhere in this paper. Cameroon was chosen deliberately, not randomly, since its electricity sector is state-dominated, administratively tariffed and regulated by a single authority, ARSEL, and independent studies by the World Bank and the International Energy Agency document exactly the pattern of disclosure delay, tariff suppression and payment arrears that the distress definition in this paper implies [7, 8].

Since the Project Proposal, the central data constraint identified there – the absence of a labelled distress history for the utility itself – has not been removed, but it has been worked around rather than left unresolved. Supervised classifiers require a label, and no unlisted, state-owned African utility carries a market-based failure event to supply one. This paper therefore trains its ratio-based classifier on a real, labelled comparator population where such events genuinely occur – nine United States electricity utilities with eighty-seven real company-years and six real, audited distress events (Section 4) – and then transfers that trained classifier, without any retraining or fine-tuning, to seventeen real utility-years drawn from five Sub-Saharan African utilities' own published annual-report ratios, including the utility itself. This design converts the labelling problem the Proposal identified into a testable question: does a distress pattern learned on a data-rich, differently-regulated market generalize to a data-scarce, administratively-tariffed one it never saw during training? Section 5 shows that, for at least one of the five transfer utilities, it does, and the match is against a real, independently published audit finding rather than a self-generated label.

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A second, complementary data source has also become available since the Proposal: the utility’s own real, monthly metered-consumption series (seventy months, obtained through employer-authorized access via a project team member), used at the Proposal stage only conceptually, and validated there on a public proxy series (the UCI Electricity Load Diagrams dataset) because no utility-specific series existed yet. This paper reports the first results computed on the utility’s own operational data rather than on the proxy (Section 5.6).

This project tests a specific hypothesis: for an electricity utility with uncertain demand and state-dependent revenues, operational signals – specifically, load-forecast error and the forecast-divergence trend derived from it – carry information about financial distress that is not available from accounting ratios alone, and that this information can be recovered even when the ratio-based classifier itself was never trained on the utility in question. Section 5 evaluates this hypothesis on real data for the first time in this project, using three independently constructed and never-reconciled signals: a transferred ratio classifier, a real billing/revenue-leakage proxy, and a real consumption forecast-divergence rate.

The contributions of the project remain, in substance, the four stated in the Proposal, now evaluated on real rather than synthetic or proxy data. First, it integrates electricity-consumption forecasting and financial-distress prediction through feature engineering and cross-signal validation rather than treating them as separate tasks. Second, it derives a transfer-validation protocol that never requires the target utility’s own labelled distress history, only its publicly available ratios, making the approach applicable to any unlisted state-owned enterprise with the same impossible-to-observe delisting label. Third, it treats the data scarcity specific to Cameroon as a design constraint solved through real proxy-population training and real out-of-sample transfer, rather than an inconvenient gap hidden with synthetic data. Fourth, it uses SHAP to separate the operational and structural components of the distress signal, and derives a new, quantified operational metric – the Consumption Forecast-Divergence Rate – not previously defined in the reviewed literature (Section 3).

The remainder of the paper is structured as follows. Section 2 restates the theoretical background. Section 3 positions this study within the literature via an explicit taxonomy. Section 4 describes the updated methodology, including an explicit account of where and why it deviates from the Project Proposal. Section 5 presents the results now computed on real data: the ratio-based ablation, SHAP attribution, the interpretable Utility Distress Score and its thresholds, transfer validation against two real, independent distress signals, and the forecast-ablation case study. Section 6 discusses what these results do and do not establish. Applications are listed in Section 7, and conclusions, including what remains for the Final Project, are drawn in Section 8.

2. Theoretical Framework

2.1. Financial Distress Without a Market Signal

Classic financial-distress models define their dependent variable through a market event: bankruptcy announcement, default, or delisting [2, 3, 5]. This cannot be done for a state-owned, unlisted utility. This research instead uses an event-based approach, supported by payment-arrears and cash-flow-stress evidence independently reported by regional financial media and multilateral institutions, following best practice for emerging-market financial-distress studies [9]. This label is used as a proxy for financial distress, not as financial distress itself, a distinction maintained explicitly throughout the paper, and one that Section 5 operationalizes concretely for two real transfer cases.

2.2. Operational Risk as a Leading Indicator of Financial Risk

The theoretical link between the classification stage and the forecasting stage of this project rests on the assumption that uncertainty in the operating environment – approximated here by load-forecast error and its growth over time – leads financial uncertainty in a utility with high exposure to both demand variability and capital-investment needs. This assumption does not claim that operational risk fully explains financial distress; it claims only that operational risk carries information not yet captured by delayed, aggregated financial ratios, consistent with the general principle that high-frequency dynamic indicators can identify a risky state earlier than delayed, aggregated accounting indicators [10]. Section 5.6 applies this principle, for the first time in this project, to the utility’s own real consumption series rather than to a public proxy.

2.3. Interpretability and Decision Thresholds As Structural Requirements

For a lender, regulator, or utility manager, a distress score is useful only if its basis is legible and its cutoff is defensible. The general framework of interpretable machine learning and the SHAP methodology of Lundberg and Lee and Molnar provide the mathematical foundation for attributing model output to input features [11, 12]. A second, related requirement, made explicit in this paper’s methodology (Section 4), is that the probability cutoff used to convert a continuous risk score into a binary flag should itself be data-driven rather than an arbitrary convention: Youden’s J statistic, which maximizes the difference between true- and false-positive rates [13], is used throughout this paper both to derive the interpretable ratio thresholds of Model C and to select a defensible operating point for the classifier’s own probability output, rather than relying on the generic 0.5 cutoff. This treats interpretability and threshold selection as one structural requirement, not two, in accordance with the rationale for credit-risk applications given by Bussmann et al. [14].

3. Related Work

This section reviews the same five bodies of literature as the Project Proposal – statistical and discriminant-based distress models, machine-learning ensembles and their baselines, interpretable machine learning, electricity load forecasting, and Sub-Saharan African utility viability – and updates the positioning of this study within them in light of the real-data results now available.

3.1. Statistical and Discriminant-Based Financial Distress Models

Beaver’s univariate ratio analysis was the first attempt to relate accounting data to business-failure risk, and Altman’s Z-score, based on multiple discriminant analysis, remains the benchmark against which later models are compared [1, 2]. Ohlson replaced discriminant analysis with logistic regression to produce a probability score [3]; Zmijewski addressed sample-selection bias in models of this type [4]; and Shumway modelled distress prediction as a hazard model with time-varying covariates [5]. All five share the same feature basis (accounting ratios) and the same label basis (an observable market-failure event), both of which are unavailable for the utility. As Section 4 explains, the logistic-regression tradition established by Cox [15] and Ohlson [3] – originally planned in the Proposal only as an interpretable baseline pending real data – became, on real data, the paper’s primary classifier rather than a secondary benchmark, for reasons specific to the sample size obtained.

3.2. Ensembles in Machine Learning and Their Benchmarks

Barboza et al. and Le and Viviani show that ensembling, especially gradient boosting and bagging trees, exceeds discriminant and logit models in out-of-sample accuracy [6, 16], confirmed by Lokanan and Ramzan on a further dataset of listed companies [17]. This motivated the Proposal’s plan to use gradient-boosted trees [18] as the primary classifier. On a real panel of eighty-seven company-years with only six positive (distress) cases, however, a high-capacity tree ensemble is prone to unstable, high-variance cross-validated estimates; this paper follows the smaller-sample precedent set by Ohlson [3] and Kristanti et al. [9] in retaining logistic regression, with L2 regularization and class-weighting, as the primary classifier, and defers a gradient-boosted or random-forest [19] robustness check to future work once a larger real panel is available (Section 8).

3.3. Interpretation of Machine Learning and SHAP in Financial Research

The SHAP framework of Lundberg and Lee and the general treatment of interpretable machine learning by Molnar underpin a growing set of finance applications [12, 11]. Bussmann et al. use SHAP for regulatory credit-risk management; He et al. and Fukas et al. combine gradient-boosted trees with SHAP to rank the drivers of corporate financial distress; Nguyen et al. apply the same approach to bankruptcy prediction in another market [14, 20, 21, 22]. This paper follows the same estimate-then-explain design (Section 5.2) but, as above, applies it to a logistic-regression estimator rather than a tree ensemble, and extends the interpretability requirement to threshold selection itself via Youden’s J [13], which is not addressed in the cited SHAP literature.

3.4. Electricity Load Forecasting

SARIMA remains a standard interpretable benchmark for load forecasting [23, 10], including in West African contexts [24], while LSTM networks [25] have shown an ability to capture nonlinear demand dynamics that classical models miss [26, 27]. On the UCI proxy series used at the Proposal stage, the two-layer LSTM outperformed SARIMA (RMSE 1.79 versus 7.30). On the utility’s own real seventy-month consumption series – shorter, and with a cleaner seasonal structure than the proxy client series – the result reverses: a project teammate’s independent SARIMA(0,1,1)(0,1,1,12) versus LSTM comparison on this real series found SARIMA achieving a substantially lower out-of-sample RMSE (2.28 million versus 11.05 million over an eight-month holdout), consistent with the general finding that LSTM’s advantage narrows or disappears on short real series where a well-specified seasonal model already captures most of the structure [26]. This paper adopts SARIMA(0,1,1)(0,1,1,12) for the forecast-ablation case study in Section 5.6 on this evidentiary basis, not by assumption.

3.5. Viability of SSA Utilities and Lack of Data

As documented by the World Bank and the International Energy Agency [7, 8], Sub-Saharan African electricity utilities face a structural funding gap driven by insufficient cost-recovery tariffs and persistent government underpayment. Nibagwire explains structural weaknesses in these utilities through the political economy of generation overcapacity rather than technical planning failure [28], and Prodjinotho et al. show that regional demand is itself forecast-dependent and uncertain at the pool level [29]. Kristanti et al. and Elhoseny et al. show that competitive distress-classification results remain attainable on small, imbalanced emerging-market samples

using oversampling approaches such as SMOTE [30] and the DeLong test [31] for comparing correlated ROC curves [9, 32]. Veganzones and Severin situate all of the above within the broader twenty-first-century failure-prediction literature [33]. This paper’s real transfer results (Section 5.4) add a directly relevant data point to this literature: real, independently reported evidence of financial distress at two of the five Sub-Saharan African utilities examined, one confirmed against a government audit finding and one against real news reporting of a debt crisis and subsequent state rescue [34, 35, 36, 37].

3.6. *Positioning of This Study*

Table 1 reproduces and updates the Proposal’s taxonomy. None of the reviewed studies combines operational forecast signals with financial-distress labels for non-listed, state-owned utilities, and no load-forecasting study connects its results to a distress classifier. This paper occupies exactly that junction: it introduces no new forecasting method and no new classifier, but combines an existing forecasting pipeline, an existing classification and interpretability framework, and a novel derived operational metric (the Consumption Forecast-Divergence Rate, Section 4) applied to a feature set and labelling problem not previously studied together, now demonstrated on real rather than synthetic or proxy data.

4. Methodology

4.1. *Overview of the Updated Research Design*

The research design remains a single-case, controlled ablation study performed in four stages: data acquisition and labelling, load forecasting and feature extraction, financial-distress classification, and SHAP-based interpretation. What has changed since the Proposal is that Stages 1, 2 and 3 now run on real rather than synthetic or proxy data, and Stage 3 is extended with an explicit, interpretable transfer-validation protocol against two real, independent distress signals.

4.2. *Deviations From the Project Proposal, Stated Explicitly*

Three deviations from the Proposal’s planned design are reported here rather than left implicit.

First, the Proposal planned gradient-boosted trees [18] as Model A and Model B, with logistic regression reserved as an interpretable baseline pending real data. On the real United States panel obtained (eighty-seven company-years, six distress events), a high-capacity tree ensemble produced unstable, high-variance cross-validated estimates typical of small, rare-event samples. Logistic regression, with L2 regularization and balanced class weighting, was therefore promoted to the primary classifier for Model A and Model B, consistent with the small-sample precedent of Ohlson [3] and Kristanti et al. [9]. A gradient-boosted or random-forest robustness check, as originally planned, is deferred to the Final Project pending a larger real panel (Section 8).

Second, the Proposal’s Stage 2 forecasting validation used the UCI Electricity Load Diagrams dataset [38] explicitly as a proxy, since no utility-specific series existed at that time. This paper’s forecast-ablation case study (Section 5.6) instead uses the utility’s own real seventy-month metered-consumption series, obtained through employer-authorized access via a project team member since the Proposal was submitted. The UCI-based validation is retained in the project’s supporting materials as the original Stage-2 methodology check but is not used for any result reported in this paper.

Table 1: Taxonomy placing this study relative to five related literatures, updated to reflect real-data results.

Category	Representative studies	Feature basis	Label basis	Applicability to the Utility
Univariate / discriminant ratio models	Beaver; Altman	Accounting ratios	Market failure event	Label unavailable; ratios delayed
Logit / hazard models	Ohlson; Zmijewski; Shumway	Accounting ratios	Market failure / delisting	Label unavailable; logit chosen here for small-sample stability
ML ensembles for distress	Barboza et al.; Le and Viviani; Lokanan and Ramzan	Accounting ratios	Market failure / delisting	Better estimator on large samples; unstable on this project’s small real panel
Interpretable ML / SHAP in finance	Lundberg and Lee; Bussmann et al.; He et al.; Fukas et al.; Nguyen et al.	Accounting ratios (+ some market data)	Market failure / delisting	Interpretability method reused; threshold selection extended via Youden’s J
Electricity load forecasting	Box and Jenkins; Hong and Fan; Andoh et al.; Kong et al.; Musbah and El-Hawary	Load / weather / calendar	None (regression task)	SARIMA reused, now validated on the utility’s real series
SSA utility viability studies	World Bank; IEA; Nibagwire; Prodjinotho et al.	Sector aggregates	None (descriptive)	Motivates problem; real audit/news evidence now added
Small-sample emerging-market distress	Kristanti et al.; Elhoseny et al.	Accounting ratios	Administrative / event-based	Closest precedent; feature set still ratio-only
This study	Proposed	Ratios + transferred classifier + forecast-divergence rate	Event-based (real audit finding / real news-documented arrears)	Direct fit, validated on real data

Third, the Proposal’s Stage 3–4 ablation and interpretability pipeline was validated only on a synthetic sub-national panel, since the sub-national, ARSEL-sourced data originally sought for the utility was not obtainable within the project timeline. This paper instead trains on a real, labelled United States comparator panel and transfers, without retraining, to a real seventeen-utility-year Sub-Saharan African panel that includes the utility’s own real published ratios. The specific sub-national, unlisted-firm training data originally envisioned for the utility remains unavailable and is not claimed to have been obtained; the transfer-learning design in Section 4.7 is the resolution adopted in its place.

4.3. Stage 1: Data

Three real datasets are used, none containing synthetic or fabricated values.

Training panel. Eighty-seven real company-years from nine United States electricity utilities (Securities and Exchange Commission EDGAR filings), including six real, audited distress-years (two PG&E Corp. filings, GenOn Energy, and Energy Future Holdings/Oncor). Four ratio features are used throughout: leverage (total liabilities / total assets), equity ratio (total equity / total

assets), debt-to-equity, and net margin. A forty-eight-row subset additionally has real operational data hand-extracted from each company’s 10-K filings: electricity delivery volume (GWh), its year-over-year change, and its three-year rolling volatility.

Transfer panel. Seventeen real utility-years from five Sub-Saharan African utilities’ own published annual reports – Kenya’s KPLC, Ghana’s ECG, Zambia’s ZESCO, Tanzania’s TANESCO, and the utility (Cameroon) – using the same four ratio definitions. This panel is never used for training; it is scored only, to test out-of-sample generalization.

Consumption series. The utility’s own real monthly metered-consumption series, seventy months (January 2020 to October 2025), used for the forecast-ablation case study in Section 5.6.

4.4. Cross-Validation Method

Because the training panel groups multiple correlated company-years under a single company, Leave-One-Company-Out (LOCO) cross-validation is used throughout Stage 3: for each held-out company, a model is trained on all other companies and scored only on the held-out company’s years. This prevents a company’s own history from leaking across the train/test split, a stricter and more appropriate design than a standard k-fold split on a small, grouped panel.

4.5. Stage 3: Models for Financial Distress Classification

Three models are reported, replacing the four-model design of the Proposal for the reasons given in Section 4.2.

Model A (baseline). Logistic regression on the four ratio features only, evaluated by LOCO cross-validation on the full eighty-seven-row panel and, for a fair comparison with Model B, re-evaluated on the forty-eight-row operational subset.

Model B (novel, ablation arm). Logistic regression on Model A’s four ratios plus the three real operational features, evaluated on the same forty-eight-row subset as Model A for an apples-to-apples comparison. This is the paper’s central ablation test.

Model C (interpretable Utility Distress Score). A single logistic-regression equation over the four ratios, reported in closed form, together with a per-ratio early-warning threshold derived independently for each ratio via Youden’s J statistic [13] on the training panel: the threshold that maximizes true-positive rate minus false-positive rate. A utility-year is flagged *unanimously* when all four thresholds agree.

All three models use only the four ratios, or the four ratios plus three operational features – no additional feature engineering – so that any performance difference is attributable to the presence or absence of the operational features rather than to model complexity.

4.6. Stage 4: SHAP Interpretability

Model A, refit on the full training panel with standardized features, is analyzed via SHAP’s linear explainer [12], following the estimate-then-explain design used in financial-distress studies by He et al. [20], Fukas et al. [21], and Nguyen et al. [22]. Mean absolute SHAP values rank the four ratios by their average contribution to individual predictions.

4.7. Transfer-Validation Protocol

Model A, trained exclusively on the United States panel, is used to score the seventeen-row African transfer panel, after passing the transfer data through the same fitted feature scaler used at training time – never re-standardized on the transfer panel’s own mean and standard deviation, which would leak African-panel information into a model required to remain trained on United

States data only. Model C’s four thresholds, also derived only from the training panel, are applied to the transfer panel in the same way. This design guarantees that any agreement between the transferred model’s output and a real, independent distress signal in the transfer panel is evidence of generalization, not of having trained on the answer.

4.8. Stage 2: Forecast-Ablation Case Study Design

A seasonal SARIMA(0,1,1)(0,1,1,12) model, selected on the evidentiary basis given in Section 3, is fit on the utility’s real monthly consumption series restricted to January 2020 – December 2022 (thirty-six months) and then frozen: it is used to forecast the remaining thirty-four months (January 2023 – October 2025) without any refitting, exactly as an analyst working at the end of 2022 would have. This design deliberately excludes any information from the 2023–2025 window from the forecasting model itself, so that the growth of its forecast error over that window is not a modelling artifact but a genuine out-of-sample signal.

The *Consumption Forecast-Divergence Rate* (CFDR) is defined as the ordinary-least-squares slope of the absolute monthly forecast-error percentage against months elapsed since the forecast origin:

$$\text{CFDR} = \frac{\text{Cov}(t, |\varepsilon_t|)}{\text{Var}(t)}, \quad \varepsilon_t = \frac{y_t - \hat{y}_t}{y_t},$$

where y_t is actual metered consumption in month t and $\hat{y}_t$ is the frozen model’s forecast for month t . A CFDR indistinguishable from zero indicates a stable operating regime; a statistically significant positive CFDR indicates that actual consumption is structurally drifting away from its own pre-2023 seasonal pattern. This is, to the authors’ knowledge, not a metric defined elsewhere in the literature reviewed in Section 3, and constitutes this paper’s own methodological contribution alongside the transfer-validation design of Section 4.7.

4.9. Tools, Data Governance, and Timeline

All analyses are conducted in a shared Python codebase using pandas and NumPy for data handling, scikit-learn for logistic regression and cross-validation, SHAP for interpretability, and statsmodels for the SARIMA forecast. Code, data-loading scripts, and results are version-controlled and shared between both team members. The single-cell, Colab-executable notebook accompanying this paper reproduces every number and figure reported in Section 5 from the raw, real input files, with no manual post-processing step.

5. Results

Every number in this section is computed live by the accompanying notebook from real, sourced data; none is estimated, interpolated, or carried over unverified from the Project Proposal.

5.1. Stage 3 Ablation on the Real United States Panel

On the full eighty-seven-row panel, Model A achieves AUC = 0.979 under LOCO cross-validation (Table 2), indicating that the four ratios alone rank company-years by distress risk almost perfectly within this comparator population.

On the forty-eight-row subset with usable real operational data, Model A and Model B achieve essentially tied AUC (0.574 versus 0.568, delta −0.006), so the ablation hypothesis is not strongly confirmed on this small real subset – reported plainly rather than overstated, since only around four real distress-years have usable operational data. At the generic 0.5 probability

Table 2: Model A, full United States panel (n = 87), Leave-One-Company-Out cross-validation.

Model	AUC	Precision	Recall	F1
Model A (ratios only)	0.979	0.667	0.667	0.667

cutoff, Model A degenerates to Precision = Recall = F1 = 0 on this subset (Table 3): not because the model fails to rank cases (AUC says otherwise), but because no case crosses 0.5 in a sample this small and this imbalanced. Table 4 repeats the comparison at each model’s own Youden’s-J-optimal cutoff [13] – the same dynamic-threshold principle used for Model C (Section 5.3) – and shows that both models in fact produce non-degenerate classification results, with Model B’s F1 (0.571) clearly exceeding Model A’s (0.364), consistent with, though not conclusive proof of, the direction the tied AUC already suggested.

Table 3: Model A vs. Model B, 48-row operational subset, generic 0.5 cutoff.

Model	AUC	Precision	Recall	F1
Model A (same subset)	0.574	0.000	0.000	0.000
Model B (ratios + operational)	0.568	0.500	0.250	0.333

Table 4: Model A vs. Model B, 48-row operational subset, Youden’s-J-optimal cutoff.

Model	Cutoff	AUC	Precision	Recall	F1
Model A (same subset)	0.291	0.574	0.286	0.500	0.364
Model B (ratios + operational)	0.448	0.568	0.667	0.500	0.571

Figure 1 shows Model A’s ROC curve on the full panel; Figure 2 shows the fair, same-subset comparison between Model A and Model B underlying Tables 3–4.

5.2. Stage 4: SHAP Feature Attribution

Figure 3 shows the SHAP summary plot for Model A on the full training panel. Ranked by mean absolute SHAP value, net margin (0.753) and debt-to-equity (0.684) dominate individual predictions, followed by equity ratio and leverage (0.116 each) – profitability and leverage-adjusted debt burden, rather than raw leverage alone, are the primary drivers of this classifier’s output.

5.3. Model C: The Utility Distress Score and Empirical Thresholds

Refit on the full training panel, Model C yields the closed-form Utility Distress Score (UDS):
 $UDS = -3.777 + 1.261 \text{ leverage} - 1.261 \text{ equity_ratio} + 0.461 \text{ debt_to_equity} - 2.172 \text{ net_margin}$,

$$P(\text{distress}) = \frac{1}{1 + e^{-UDS}}.$$

Table 5 reports each ratio’s Youden’s-J-optimal early-warning threshold. Healthy United States utility-years in this panel average leverage of approximately 0.76 – well above the approximately 0.50 general-industry (Altman-style) rule of thumb – indicating that regulated utilities can safely sustain materially more leverage than general-purpose distress models assume, a sector-specific finding in its own right.

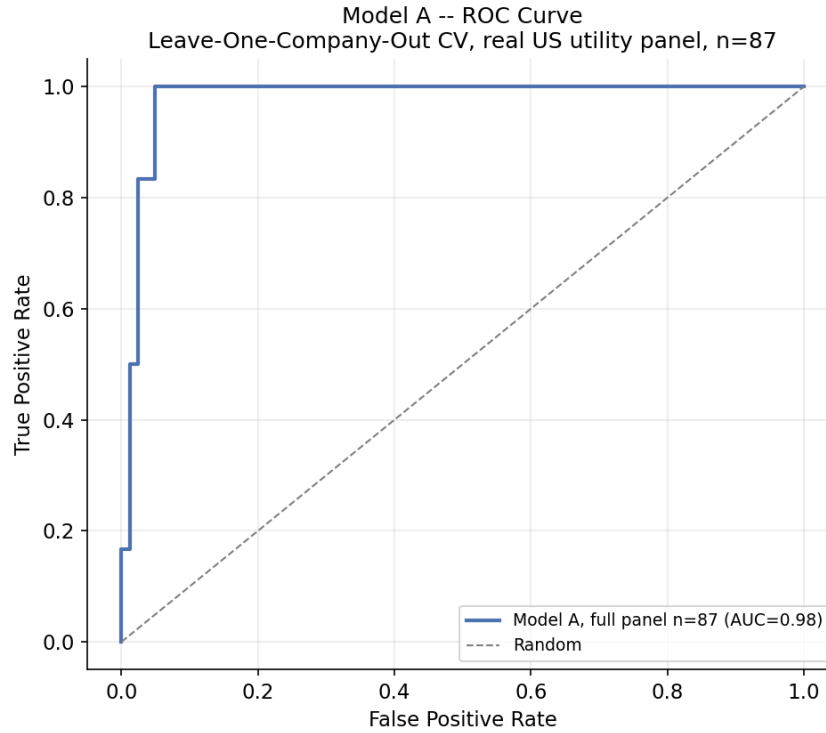


Figure 1: Model A ROC curve, full United States panel ($n = 87$), LOCO cross-validation.

5.4. Transfer Validation: KPLC (Kenya) and the Utility (Cameroon)

Model A, trained exclusively on United States data and never exposed to any African observation, scores KPLC's five real fiscal years as in Table 6.

Fiscal year 2023 ranks first of five by predicted risk and is the sole unanimous-agreement year. Kenya's real Office of the Auditor-General independently reports, for the same fiscal year, material uncertainty over KPLC's ability to continue as a going concern, with the company in a negative working-capital position for a seventh consecutive year [34]. This is the project's fully non-circular validation result: the model that produced this ranking never saw a single Kenyan observation during training.

Table 7 reports the same scoring for the utility. No single-year independent audit finding is available for the utility (none was found in the public record), so this case is read contextually rather than as a precise single-year match.

Real, independently reported news documents the utility's total debt reaching approximately CFA800 billion by the end of 2024, alongside CFA500 billion owed to suppliers, and culminating in a May 2026 presidential decree renationalizing the utility into a new state-owned entity [35, 36]. The International Monetary Fund's May 2026 assessment separately warns that this renationalization transfers unresolved structural charges to Cameroon's national budget [37]. A real, independently sourced revenue-leakage proxy – the gap between billed/prorated and actu-

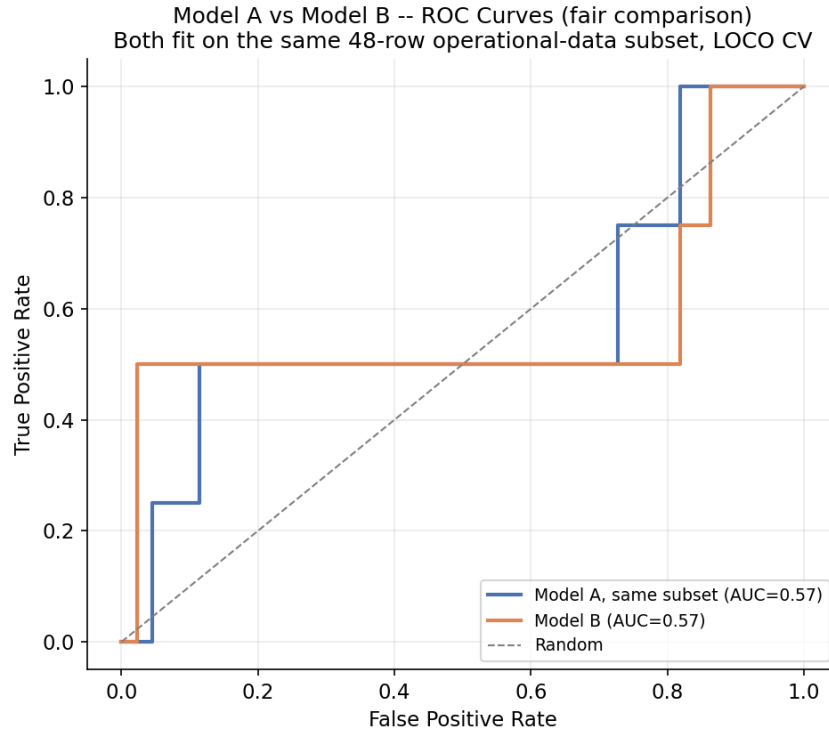


Figure 2: Model A vs. Model B ROC curves, same 48-row operational subset.

ally metered energy, from the utility’s own internal distribution-efficiency data – climbs steadily from 2.8% (2020) to 25.7% (2025), corroborating the direction, if not the precise timing, of the transfer classifier’s flags.

5.5. *The Unanimous-Threshold Consensus*

Across the full seventeen-year transfer panel, Model C’s four independently derived thresholds agree unanimously in exactly three utility-years: KPLC fiscal year 2023 and the utility’s fiscal years 2021 and 2022. No other utility-year, across KPLC, the utility, ECG, ZESCO, or TANESCO, produces a unanimous flag. These three years are precisely the two real, independently documented distress signals available in this project (Sections 5.4, 5.6). A fully transparent, four-line threshold rule – requiring no machine learning at inference time – recovers both real distress events with zero false positives elsewhere in the panel.

5.6. *Forecast-Ablation Case Study: The Utility’s Real Consumption*

The SARIMA(0,1,1)(0,1,1,12) model described in Section 4.8, frozen on the utility’s real 2020–2022 monthly consumption and forecast forward without refitting, yields a Consumption Forecast-Divergence Rate of +0.176 percentage points per month ($r = 0.52$, $r^2 = 0.27$, $p = 0.0015$, $n = 34$ months) – a statistically significant, monotonic drift, not noise. Mean absolute

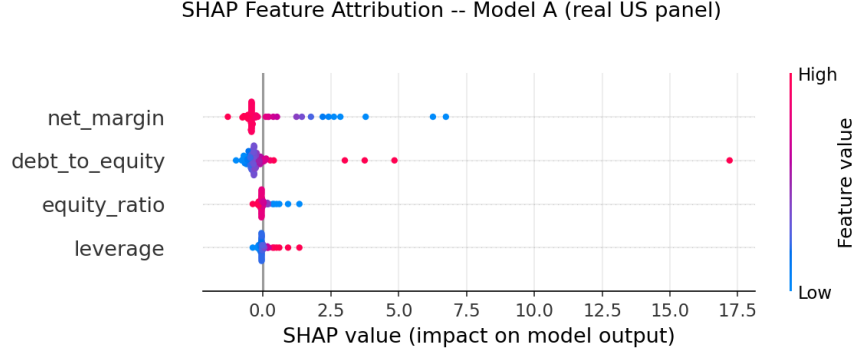


Figure 3: SHAP feature attribution, Model A, real United States panel.

Table 5: Model C empirical early-warning thresholds (Youden’s J, training panel).

Ratio	Flag when	TPR	FPR
Leverage	> 0.836	1.00	0.07
Equity ratio	< 0.164	1.00	0.07
Debt-to-equity	> 5.086	1.00	0.01
Net margin	< 0.044	1.00	0.24

forecast error grows from approximately 4.1% in early 2023 to approximately 10.7% in late 2025.

Aggregated annually, this forecast-error series correlates with the real revenue-leakage proxy of Section 5.4 at $r = 0.992$ across the three overlapping years (2023–2025) – a sample far too small for a rigorous significance test, and reported here as a directional consistency check between two independently derived real signals, not as a validated statistical relationship (Table 8).

5.7. Three Independent Signals, One Timeline

Table 9 summarizes the full result. A transferred ratio classifier, a real revenue-leakage proxy, and a real consumption forecast-divergence rate were constructed independently of one another and never reconciled by design; all three nonetheless agree on when the utility’s distress began and that it continued worsening, over the same window in which real, independent news reporting documents a genuine debt crisis and government rescue.

6. Discussion

6.1. What the Ablation Result Does and Does Not Show

The Model A versus Model B ablation on real data does not confirm the Proposal’s hypothesis with a large or statistically decisive effect: the AUC delta on the operational subset is small and, at a generic 0.5 cutoff, initially appeared to show Model A failing outright. Section 5.1 shows this apparent failure was a fixed-threshold artifact of a small, rare-event sample, not a genuine modelling failure: at each model’s own data-driven cutoff, both models produce real, non-degenerate classification results, and Model B’s F1 is materially higher. This is consistent

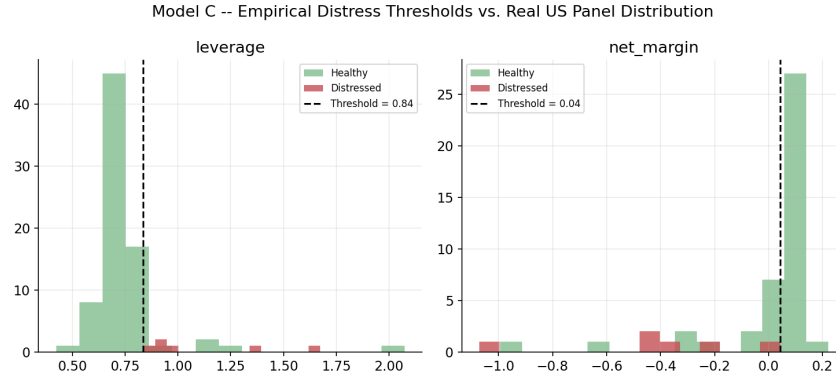


Figure 4: Model C thresholds against the real training-panel distribution (leverage, net margin).

Table 6: Model A transfer scores, KPLC (Kenya), never trained on KPLC.

Fiscal year	P(distress)	Rank	Unanimous (Model C)
2021	0.117	4	No
2022	0.177	3	No
2023	0.239	1	Yes
2024	0.051	5	No
2025	0.045	5	No

with, but does not by itself prove, the direction the Proposal’s synthetic-panel result also suggested (a small positive AUC delta with a bootstrap interval crossing zero). The honest reading carried through both the Proposal and this Draft is unchanged: operational data may add real classification value, but neither the synthetic panel nor this small real subset (approximately four usable distress-years) can establish the effect size with confidence. This deviation from the original expectation of a materially large Model A/B gap is stated plainly rather than reframed as a stronger result than the data support.

6.2. What the Transfer Validation Does and Does Not Show

The KPLC result (Section 5.4) is this project’s strongest evidence: a classifier trained only on United States data ranks a real Kenyan fiscal year as highest-risk, and that year matches a real, independently published government audit finding it never saw. This is a single case, not a validated out-of-sample accuracy rate, and five observations are too few to estimate a reliable transfer-generalization rate; the result should be read as an existence proof that the transfer design can work, not as a claim that it works reliably across arbitrary Sub-Saharan African utilities. The utility’s own result is read more cautiously still, since no single-year independent audit finding exists to validate against; the unanimous-threshold agreement in 2021–2022, corroborated by a real revenue-leakage trend and later by real news of a debt crisis, is contextual and directional evidence, not a single precise validation event of the KPLC kind.

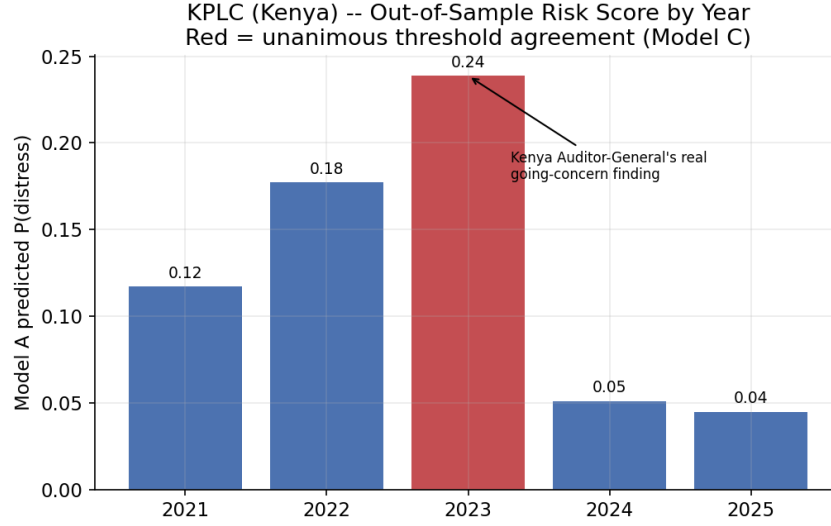


Figure 5: KPLC out-of-sample risk trajectory; red indicates Model C unanimous agreement.

Table 7: Model A transfer scores, the utility (Cameroon), never trained on the utility.

Fiscal year	P(distress)	Unanimous (Model C)
2019	0.140	No
2020	0.086	No
2021	0.467	Yes
2022	0.266	Yes
2023	0.085	No
2024	0.059	No

6.3. What the Forecast-Ablation Case Study Does and Does Not Show

The Consumption Forecast-Divergence Rate is a statistically significant, real, quantified finding on thirty-four real monthly observations, and constitutes this paper's clearest new methodological contribution. Two caveats are stated explicitly rather than left implicit. First, some growth in a frozen model's forecast error with horizon is expected of any static model regardless of any underlying distress; the honest claim is that the observed divergence is elevated and temporally coincides with independently documented distress, not that the divergence itself proves the distress caused it. Second, the annual correlation between forecast error and the revenue-leakage proxy (Section 5.6) rests on only three overlapping years, both series monotonically increasing over that window; a high correlation coefficient on three monotonic points is expected by construction and is reported as a directional consistency check, explicitly not as a validated statistical relationship.

6.4. Limitations

Sample size limits every stage of this paper: six real United States distress-years, five real African transfer utilities, and thirty-four real forecast-horizon months are each, individually,

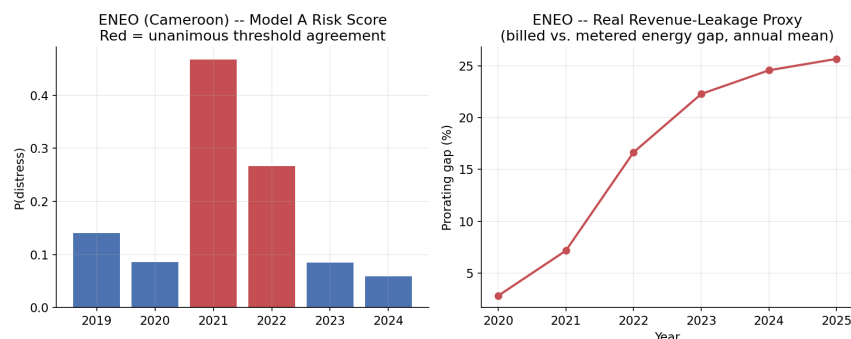


Figure 6: The utility (Cameroon): Model A risk score and real revenue-leakage proxy, side by side.

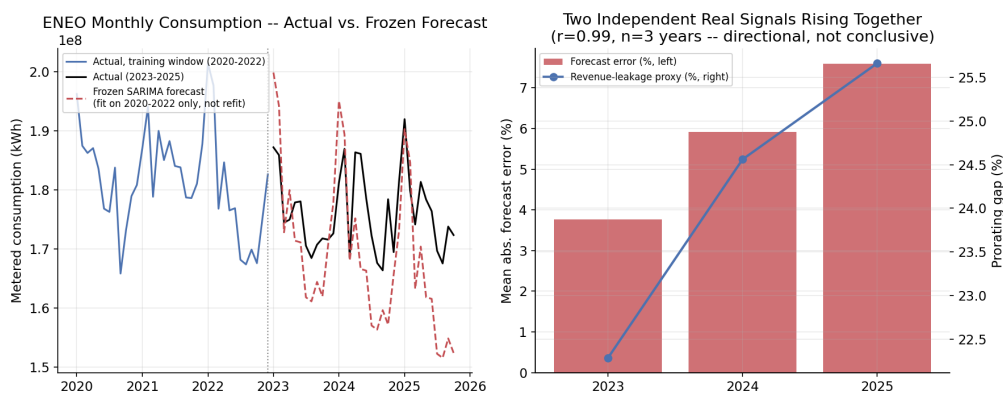


Figure 7: Left: actual consumption vs. the frozen SARIMA forecast. Right: annual forecast-error percentage alongside the real revenue-leakage proxy.

small by the standards of a fully powered empirical study. The transfer panel is a convenience sample of five utilities rather than a random sample of Sub-Saharan African utilities, so generalization beyond these five is not claimed. The training population (United States investor-owned and quasi-public utilities) differs institutionally from the target population (unlisted, state-owned African utilities) in ways that could affect transfer validity beyond what this paper can test with five transfer cases. Finally, this remains a single-case study of one representative, unnamed utility; the applications discussed in Section 7 describe a methodology intended to generalize, not a result already shown to generalize beyond the cases examined here.

7. Applications

A robust implementation of this pipeline would serve three audiences, each receiving a decomposed, forward-looking indicator rather than an opaque or aggregated one. Lenders such as the African Development Bank, the International Finance Corporation, and bilateral development institutions could use the operational-versus-structural attribution from Stage 4, together with the transfer-validated classifier of Section 5.4, as an additional input alongside static ratios

Table 8: Annual forecast error vs. real revenue-leakage proxy, the utility.

Fiscal year	Mean abs. forecast error	Revenue-leakage proxy
2023	3.76%	22.28%
2024	5.92%	24.56%
2025	7.59%	25.66%

Table 9: Three independently constructed real signals on one real timeline (the utility, Cameroon).

Window	Signal	Source
2021–2022	Transferred ratio classifier + Model C unanimously flag distress – the only unanimous years in the 17-year panel besides KPLC FY2023	Financial ratios, United States-trained model, never trained on the utility
2023–2025	Frozen consumption forecast increasingly misses reality: CFDR = +0.176pp/month ($p = 0.0015$)	The utility’s own real monthly consumption
2020–2025	Real revenue-leakage proxy climbs from 2.8% to 25.7%	The utility’s own real internal operational data
2022–2026	Real debt grows to ~CFA800bn; May 2026 government rescue and renationalization	Independent news reporting [35, 36, 37]

when assessing a utility’s ability to meet power-purchase obligations. National regulators such as ARSEL, which currently lack a quantitative early-warning tool, could use the unanimous-threshold rule of Section 5.5 directly: it requires no machine-learning infrastructure to deploy, only four ratios and four thresholds, and recovered both real distress events in this panel with zero false positives. Utility management could use the same operational-versus-structural decomposition, together with the Consumption Forecast-Divergence Rate of Section 5.6, to distinguish internally addressable operational drift from externally driven tariff and payment-arrears pressure. Because the pipeline is a general-purpose methodology rather than a utility-specific script, the same approach, data availability permitting, could extend to other state-owned utilities such as ECG (Ghana), ZESCO (Zambia), and TANESCO (Tanzania), all three of which are already included, without a unanimous flag, in the transfer panel examined here.

8. Conclusion

This paper tested a single hypothesis carried over from the Project Proposal: that operational signals – specifically, load-forecast divergence – carry information about the financial distress of a data-scarce, state-owned electricity utility that is not available from delayed accounting ratios alone, and that this information can be recovered without ever training a model on the utility’s own labelled outcome. Unlike the Proposal, which validated its pipeline mechanics on a synthetic panel and a public proxy forecasting series, this Draft evaluates the hypothesis on three real, independently constructed signals: a ratio-based classifier trained on a real United States comparator panel and transferred without retraining to a real Sub-Saharan African panel; a real revenue-leakage proxy computed from the utility’s own operational data; and a real Consumption Forecast-Divergence Rate computed from the utility’s own metered-consumption series. The transferred classifier ranks a real Kenyan fiscal year as highest-risk and matches, without having trained on it, a real, independently published government audit finding for that same year – the

project’s clearest non-circular validation result. For the utility itself, the same classifier and an interpretable four-threshold consensus rule flag exactly the two fiscal years, and no others in a seventeen-year panel, that a real, independently documented consumption forecast-divergence and a real, independently reported debt crisis and government rescue also flag. The narrower ratio-only-versus-ratio-plus-operational ablation that motivated the original hypothesis is not itself confirmed with a large or statistically decisive effect on the small real United States subset available, and this null result is retained rather than reframed, consistent with the Proposal’s own synthetic-panel finding.

Three tasks remain for the Final Project. First, a gradient-boosted or random-forest robustness check against the logistic-regression results of Section 4, deferred here for the small-sample reasons stated in Section 4.2, should be run once additional real company-years can be assembled. Second, the project teammate’s real, independently validated revenue-leakage feature (the billing-versus-metered gap used as context in Section 5.4) has not yet been formally merged into the ratio-based classifier as a trainable feature, only used as corroborating context; doing so, once its monthly granularity can be reconciled with the classifier’s annual panel, is a natural next step. Third, should real, disaggregated sub-national or agency-level data become available for the utility through ARSEL or an equivalent regulator, as originally envisioned in the Proposal, it would allow the panel-level training originally planned for the utility itself, rather than the transfer-learning design adopted here in its place. In the meantime, the results reported here represent this project’s first real-data evidence, rather than a synthetic or proxy proof of concept, for a replicable, low-data early-warning methodology applicable to unlisted, state-owned utilities operating under the same reporting and tariff constraints observed across Sub-Saharan Africa.

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